

DEEDY: A Thai-Centric Multi-Agent Framework for Social-Media Marketing Campaign Analysis

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Abstract—Social-media marketing campaigns generate large volumes of fast-moving, context-dependent consumer reactions that are difficult to interpret with sentiment classification alone. This paper presents DEEDY (Digital Environment for Emergent Dynamics and Yield), a Thai-centric application and experimental testbed for analyzing campaign response with large-language-model (LLM) agents. For each campaign post, an Apify-based pipeline requests up to 1,000 publicly accessible comments from Facebook, Instagram, or TikTok, preserves post and collection provenance, and produces synchronized raw and Thai-aware analysis views. Campaign facts and approved context form a reality seed, whereas observed reactions reserved for evaluation remain hidden from the agents. DEEDY adapts MiroFish, an open-source multi-agent prediction engine that converts seed documents into a GraphRAG-grounded world, generates actors, runs OASIS social interactions, and supports report generation and post-run inspection. The resulting application lets marketers examine how message framing, audience composition, and platform mechanisms may shape simulated reaction, diffusion, and narrative formation. We propose leakage-resistant evaluation against held-out Thai comments using sentiment and stance distribution distance, narrative coverage, response diversity, propagation error, repeated runs, and native-Thai human assessment. OpenRouter generation completed 1,300 calls across DeepSeek V4 Flash 0731 and Qwen3 8B. An independent GLM-5.3-Flash judge then completed 1,482 blinded orientations over 741 real-synthetic pairs. It preferred the authentic comment in 59.3% of judgments, although synthetic reactions received higher mean naturalness and campaign-relevance scores. The results therefore establish neither behavioral fidelity nor a message-clarity advantage. DEEDY is positioned as an exploratory campaign-planning and social-listening instrument, not as a deterministic forecast of consumer behavior or a replacement for field research.

Index Terms—LLM agents, social-media marketing, campaign analysis, social simulation, Thai NLP, GraphRAG, MiroFish, Apify

I. INTRODUCTION

Marketing campaigns on social media are not one-way messages. Consumers reply to a campaign post, react to one another, repeat or contest emerging claims, and carry interpretations across platform feeds. A favorable first response may be displaced by complaints about price, product quality, brand conduct, or the credibility of an influencer; conversely, a small positive narrative may grow through recommendation and peer endorsement. Counting likes or assigning sentiment to comments can summarize a snapshot, but it cannot by itself represent the interaction process through which campaign narratives emerge. Agent-based modeling (ABM) provides

a bottom-up way to represent local exposure and response decisions and observe their collective consequences [1].

LLM agents extend ABM by allowing simulated actors to condition open-ended actions on a profile, memory, retrieved evidence, feed context, and earlier conversation. Generative Agents demonstrated memory-driven believable behavior [2], while Social Simulacra showed how generated users and conversations can support social-computing prototypes [3]. Systems including S^3 , OASIS, and GenSim subsequently modeled social networks, recommendation, and multiple online actions at larger scales [4], [5], [6]. These capabilities suggest a campaign-analysis application in which marketers can rehearse alternative message framing, audience composition, and feed conditions before interpreting the resulting differences as hypotheses for further testing.

MiroFish provides the application foundation for this work. The open-source project describes itself as a multi-agent prediction engine that constructs a parallel digital world from real-world seed material such as news, policy drafts, or analysis reports [7]. Its workflow has five stages: graph building with seed extraction and GraphRAG memory injection; environment setup with entity, persona, and agent construction; dual-platform social simulation with temporal memory updates; report generation; and deep interaction with individual agents or the report agent. OASIS supplies the underlying social-simulation engine. MiroFish therefore offers a useful bridge from campaign documents to an inspectable simulated environment, but its outputs must still be validated before they are used to support marketing decisions.

Applying this workflow to Thai campaigns introduces two challenges. First, Thai localization is more than translating an English prompt. Thai has no obligatory whitespace between words, and campaign comments frequently combine informal spelling, repeated characters, emoji, hashtags, transliterated English, regional language, sarcasm, indirect disagreement, and culturally specific politeness. Thai NLP resources and models have improved substantially [8], [9], [10], [11], but cultural capability remains a distinct evaluation target [12]. Second, social data collection is platform-dependent and incomplete. A reproducible study must record which campaign post was queried, the collection time, the requested limit, the returned count, and any access or visibility limitation.

A further methodological problem is that plausible campaign comments do not prove behavioral fidelity. Demographically

conditioned LLM samples can approximate some human distributions [13], yet recent work finds behavioral inconsistency across settings [14]. Operational validation against authentic comments also shows that matching surface form can be separated from matching content [15]. For this reason, campaign context needed to construct the simulation must be separated from real consumer reactions used to test it, and stochastic conditions must be repeated rather than judged from a single persuasive-looking run.

This paper presents **DEEDY**, a Thai-centric campaign-analysis application that combines an Apify data-preparation pipeline, the DEEDY Simulation Core, and an explicit validation layer. Analysts provide campaign topics and public post URLs from Facebook, Instagram, or TikTok. DEEDY requests up to 1,000 comments for each post, preserves provenance and conversation structure, and separates campaign context from held-out consumer reactions. Within the simulator, an analyst can compare a baseline campaign with counterfactual message framing, actor composition, network structure, or recommendation policy. The application reports where simulated and observed discourse agree or diverge; it does not claim to forecast the choices of a specific person or the Thai market as a whole.

Our contributions are threefold:

- We define an end-to-end Thai campaign-data pipeline that uses Apify Actors to request up to 1,000 publicly accessible comments per campaign post, normalize cross-platform fields, preserve provenance, and produce Thai-aware views for analysis and simulation.
- We adapt the MiroFish/OASIS workflow to social-media marketing through campaign-grounded actor construction, Thai linguistic attributes, configurable campaign interventions, and auditable simulation outputs.
- We specify a leakage-resistant protocol that evaluates content, population, and dynamic correspondence against held-out Thai campaign reactions using repeated runs, native-Thai review, ablations, and uncertainty reporting.

II. RELATED WORK

Generative Agents introduced memory, reflection, and planning for believable behavior [2]; S^3 , OASIS, and GenSim extended this idea to networked social environments, multiple online actions, and reusable simulations [4], [5], [6]. MiroFish connects a reality seed to graph-grounded personas, social simulation, and report generation [7]. DEEDY adapts this workflow to Thai campaign analysis and adds an explicit separation between evidence used to construct a scenario and evidence used to evaluate it.

This separation is important because plausible language is not equivalent to behavioral validity. Algorithmic fidelity compares generated and human population distributions [13], while conditioned-comment prediction directly compares generated and authentic responses to the same stimulus [15]. Generative ABMs should therefore be validated incrementally before counterfactual claims are made [16].

Thai social-media analysis adds informal spelling, code-mixing, emoji, sarcasm, indirect stance, and culturally specific politeness. Thai NLP tools and pretrained models provide useful linguistic infrastructure [8], [9], [10], [11], but cultural knowledge remains a separate capability [12]. Conversation context is also necessary because sarcasm can reverse literal polarity [17]. These findings motivate evaluation of naturalness, cultural fit, stance, and authenticity rather than sentiment alone.

III. DEEDY SYSTEM ARCHITECTURE AND WORKFLOW

A. Evidence Preparation

Public campaign posts and comments are collected from Facebook, Instagram, and TikTok through recorded Apify queries [18]. Text is preserved with emoji, Thai-English code-mixing, repeated characters, and thread cues because these may carry pragmatic meaning. Direct identifiers are removed from research views, while campaign, platform, post, time, and collection provenance are retained. The resulting corpus is not treated as a random sample of Thai consumers: platform visibility, ranking, moderation, deletion, and source selection can affect which comments are observable.

For campaign e , evidence is partitioned before generation as

$$D_e = S_e \dot{\cup} G_e, \quad (1)$$

where S_e contains only facts and context permitted to construct the scenario, and G_e contains authentic reactions reserved for evaluation. Comments, their summaries, and derived sentiment proportions from G_e are not supplied to agents. This design prevents the simulator from repeating the answers against which it will later be scored.

B. DEEDY Simulation Core

Figure 1 presents the conceptual architecture. The deployed component is called the *DEEDY Simulation Core*; MiroFish is the underlying engine. Its hybrid-agent and dual-memory design is informed by the ISAL-NLP architecture [19]. First, campaign facts are converted into a graph-grounded world containing entities, relations, and retrievable memories. Audience roles are then sampled with interests, prior attitudes, activity, language register, and uncertainty. Unsupported details remain explicit assumptions rather than facts.

The population combines a smaller cognitive tier with a larger stochastic tier. Cognitive agents retrieve relevant memories, observe a bounded feed, update their private stance, and generate a Thai action or remain silent. Stochastic agents preserve population scale using a probabilistic response conditioned on prior attitude, exposure, neighbors, and the preceding social signal. For an exposed stochastic agent i at time t ,

$$\Pr(a_{i,t} \neq \emptyset) = \sigma(\beta_0 + \beta_b b_i + \beta_d^\top \mathbf{d}_i + \beta_s s_{t-1} + \beta_x x_{i,t}). \quad (2)$$

The action space includes like, share, comment, reply, and silence. Initial exposure can be chronological, interest-based, popularity-based, or seeded through influential agents; later exposure propagates through the configured network. These

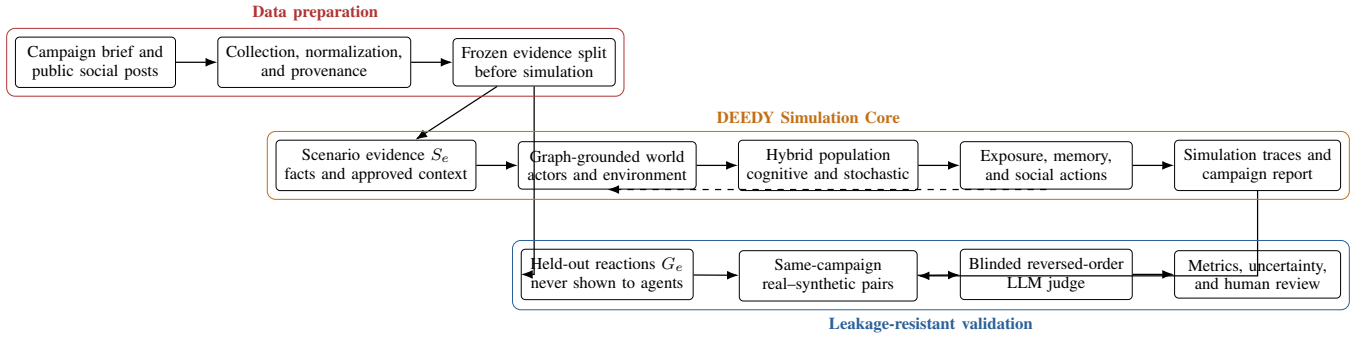


Fig. 1. DEEDY architecture and executed evaluation workflow. Campaign facts construct the simulated world, while authentic reactions remain hidden until outputs are scored. The dashed arrow represents memory feedback within the simulation.

mechanisms are experimental assumptions, not reconstructions of a platform’s proprietary recommender.

Short-term memory stores current exposures and interactions, while long-term retrieval preserves scenario knowledge and reflected experience. Every output connects an action to its actor, exposure, retrieved evidence, parent action, and simulated time. The final report summarizes sentiment, stance, narratives, reach, engagement, influential actors, and disagreement across repeated runs. This traceability supports error analysis and counterfactual comparison but does not make a simulated difference a real-world causal effect.

IV. EVALUATION AND EXPERIMENTAL RESULTS

A. Data and Experimental Design

The evaluation covers five Thai campaigns: Parameter Gelato, KFC Bucket Ware, MK Buffet, Walk/Run for Glasses by Top Charoen, and Thai Chuai Thai Plus. The corpus contains 5,000 authentic comments from 33 posts, with 1,000 comments per campaign. Aggregate sentiment analyses were used only after generation. The available crawl contained no reply links, so observed cascade correspondence could not be evaluated; network results are reported only as model sensitivity.

Experiments 1–3 test grounding, Thai prompting, and the response model. Experiment 4 varies feed and network assumptions and generates agent reactions. Experiment 5 compares a baseline campaign packet with a clarity frame that makes conditions and uncertainties more explicit. Ten random seeds are used per campaign-condition combination. Jensen–Shannon distance (JSD) measures sentiment-distribution divergence, while text quality is assessed with diversity, semantic correspondence [20], and blinded judgment. Table I reports only the conclusions supported by the executed pilot.

Across Experiments 1–5, 1,300 generation calls completed. Experiments 4–5 produced 6,400 agent records from 800 calls. Of these, 741 calls contained at least one visible reaction; 59 all-silence calls were retained in action-rate analysis but excluded from text comparison. The Experiment 4 panel showed only modest changes across feed/network prompt descriptions: DeepSeek JSD ranged from 0.2995 to 0.3233 and Qwen from 0.3274 to 0.3432. In the separate non-generative mechanism sweep, the assumed network produced substantially

larger reach and action-rate changes than population size. Because the corpus did not preserve reply edges, neither topology can be interpreted as an observed Thai interaction network.

B. Blinded LLM-as-a-Judge Evaluation

Each eligible synthetic reaction was paired with a distinct authentic comment from the same campaign. Direct identifiers and contact patterns were removed; the judge received only the campaign facts and two unlabeled texts. Each of the 741 pairs was evaluated twice with A/B order reversed, producing 1,482 valid judgments. GLM-5.3-Flash, accessed through OpenRouter [21], scored Thai naturalness, campaign relevance, social-media plausibility, pragmatic/cultural fit, contextual specificity, and unsupported-claim risk on five-point scales. It also predicted sentiment and selected which text appeared more authentic. Pair reversal was used because LLM judges may exhibit presentation and position biases [22]. Aggregate sentiment references were hidden until all judgments were complete.

The authentic comment was preferred in 59.3% of judgments, the synthetic reaction in 40.6%, and a tie in 0.1%. Treating a tie as half a win gives a synthetic-realism score of 0.406 (bootstrap 95% CI 0.374–0.438), below the 0.5 indifference point. Reversed-order preference agreed for 83.5% of pairs, while candidate A was selected in 55.9% of non-ties, confirming measurable position sensitivity.

Table II reveals a useful tension. Synthetic reactions were rated more natural, relevant, culturally appropriate, and specific, yet authentic comments were more often selected as real. Social plausibility and unsupported-claim risk showed no clear difference. The judge-labeled sentiment distribution diverged less from the aggregate reference for authentic comments (JSD = 0.077) than for synthetic reactions (0.215), reinforcing the under-production of negative response. KFC was the only campaign with a synthetic-realism score slightly above 0.5, whereas Top Charoen had the largest authenticity gap. Campaign slices are diagnostic because sample eligibility and judge priors may differ.

C. Campaign-Level Results

Table III reports all five campaign slices. The realism interval exceeded neither the 0.5 indifference point nor its uncertainty

TABLE I

EXECUTED EXPERIMENTS AND ESSENTIAL FINDINGS. JSD COMPARISONS USE THE AVAILABLE AGGREGATE REFERENCE AND ARE DIAGNOSTICS RATHER THAN HELD-OUT CAUSAL ESTIMATES.

Experiment	Comparison	Essential result
1	Grounded vs. ungrounded generation	Paired JSD intervals spanned zero for both generators; grounding did not establish a fidelity advantage.
2	Thai-context vs. translated or normalized prompts	JSD intervals spanned zero. Thai-script frequency was high but cannot establish naturalness or cultural competence.
3	Two LLMs vs. stochastic ABM	No pairwise sentiment-JSD interval excluded zero; the ABM generated states rather than language.
4	Feed and network sensitivity; LLM reaction panel	Results were much more sensitive to the assumed network than population size. The best LLM-panel mean JSD was 0.2995, and negative reactions were under-produced.
5	Baseline vs. clarity frame	Across 100 matched generator-campaign-seed pairs, clarity changed JSD by -0.0030 (95% empirical interval $-0.1888-0.1261$); no reliable advantage was found.

TABLE II

BLINDED JUDGE RESULTS OVER 741 REAL-SYNTHETIC PAIRS. SCORE GAPS ARE SYNTHETIC MINUS REAL; RISK IS THE ONLY DIMENSION WHERE HIGHER IS WORSE.

Dimension	Real	Synthetic	Gap (95% CI)
Thai naturalness	3.77	4.19	0.42 [0.29, 0.57]
Campaign relevance	2.83	4.46	1.64 [1.51, 1.77]
Social plausibility	3.84	3.88	0.04 [-0.11, 0.20]
Cultural fit	3.57	4.07	0.50 [0.35, 0.65]
Context specificity	2.23	3.14	0.91 [0.76, 1.05]
Unsupported-claim risk	1.83	1.87	0.04 [-0.06, 0.14]

for any campaign. KFC was closest to indifference, while Top Charoen was most readily distinguished from authentic discussion. Sentiment correspondence did not always move with authenticity: Top Charoen’s synthetic sentiment distribution was close to the aggregate reference even though its generated comments were rarely preferred as real. Distribution matching and comment-level realism therefore measure different properties and should be reported together.

V. DISCUSSION, LIMITATIONS, AND CONCLUSION

The results do not simply make DEEDY appear more accurate. They expose a "too polished and too relevant" failure mode: prompted agents produce coherent standalone responses, whereas authentic threads contain fragments, tagging, repetition, jokes, and side conversations. Optimizing only rubric scores could therefore reduce realism. Experiment 4 evaluates sensitivity to specified feed and network conditions, not reconstruction of platform exposure. Experiment 5 evaluates reactions to hypothetical framing, not a causal effect on consumers. The simulations are consequently most useful for scenario exploration and failure discovery.

The study has four principal limitations. First, manually selected public posts form a convenience sample rather than a representative Thai population. Second, missing reply edges prevent validation of propagation and cascade structure. Third, the sentiment reference is an aggregate weak annotation, not an independent human gold standard. Fourth, a single external LLM judge can share model priors with the generators and showed position sensitivity. Sanitization reduces but cannot eliminate privacy risk when authentic text is sent to an external service. Future evaluation should use a pre-registered campaign-level holdout, reply- and time-preserving traces, a

Thai-specialized generator, multiple judges, and blinded native-Thai raters with reported agreement.

DEEDY contributes a leakage-resistant architecture for connecting grounded Thai campaign scenarios, hybrid multi-agent simulation, and authentic-response evaluation. The completed pilot demonstrates end-to-end execution and reveals where the model departs from observed discourse; it does not establish market forecasting, campaign effectiveness, or behavioral fidelity. This distinction is central to using generative simulation as an auditable hypothesis tool rather than a substitute for field research.

REFERENCES

- [1] C. M. Macal and M. J. North, "Tutorial on agent-based modelling and simulation," in *Proceedings of the 2010 Winter Simulation Conference*, 2010, pp. 151–162.
- [2] J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–22.
- [3] J. S. Park, L. Popowski, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Social simulacra: Creating populated prototypes for social computing systems," in *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, 2022, pp. 1–18.
- [4] C. Gao, X. Lan, Z. Lu, J. Mao, J. Piao, H. Wang, D. Jin, and Y. Li, "S³: Social-Network Simulation System with Large Language Model-Empowered Agents," 2023.
- [5] Z. Yang, Z. Zhang, Z. Zheng, Y. Jiang, Z. Gan, Z. Wang, Z. Ling, J. Chen, M. Ma, B. Dong, P. Gupta, S. Hu, Z. Yin, G. Li, X. Jia, L. Wang, B. Ghanem, H. Lu, C. Lu, W. Ouyang, Y. Qiao, P. Torr, and J. Shao, "OASIS: Open agent social interaction simulations with one million agents," in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.11581>
- [6] J. Tang, H. Gao, X. Pan, L. Wang, H. Tan, D. Gao, Y. Chen, X. Chen, Y. Lin, Y. Li, B. Ding, J. Zhou, J. Wang, and J.-R. Wen, "GenSim: A general social simulation platform with large language model based agents," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*, 2025, pp. 143–150. [Online]. Available: <https://aclanthology.org/2025.naacl-demo.15/>
- [7] MiroFish Contributors, "MiroFish: A simple and universal swarm intelligence engine," Open-source software repository, 2026, accessed 12 August 2026. [Online]. Available: <https://github.com/666ghj/MiroFish>
- [8] R. Arreerard, S. Mander, and S. Piao, "Survey on Thai NLP language resources and tools," in *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. European Language Resources Association, 2022, pp. 6495–6505. [Online]. Available: <https://aclanthology.org/2022.lrec-1.697/>
- [9] W. Phatthiyaphaibun, K. Chaovavanich, C. Polpanumas, A. Suriyawongkul, L. Lowphansirikul, P. Chormai, P. Limkonchotiwat, T. Suntorntip, and C. Udomcharoenchaikit, "PyThaiNLP: Thai natural language processing in python," in *Proceedings of the 3rd*

TABLE III

CAMPAIGN-LEVEL JUDGE RESULTS. REALISM EQUALS SYNTHETIC PREFERENCE PLUS HALF THE TIE SHARE. SENTIMENT JSD COMPARES JUDGE LABELS WITH EACH CAMPAIGN’S AGGREGATE REFERENCE; LOWER IS BETTER.

Campaign	Pairs	Authentic preferred	Synthetic realism (95% CI)	Authentic JSD	Synthetic JSD
Parameter Gelato	155	0.568	0.431 [0.363, 0.498]	0.181	0.428
KFC Bucket Ware	142	0.489	0.511 [0.440, 0.585]	0.288	0.274
MK Buffet	149	0.550	0.450 [0.376, 0.527]	0.123	0.220
Walk/Run for Glasses (Top Charoen)	155	0.752	0.247 [0.182, 0.313]	0.068	0.049
Thai Chuai Thai Plus	140	0.596	0.404 [0.332, 0.475]	0.174	0.376

- Workshop for Natural Language Processing Open Source Software*, 2023, pp. 25–36. [Online]. Available: <https://aclanthology.org/2023.nlpss-1.4/>
- [10] L. Lowphansirikul, C. Polpanumas, A. Rutherford, and S. Nutanong, “WangchanBERTa: Pretraining transformer-based thai language models,” 2021.
- [11] K. Pipatanakul, P. Manakul, N. Nitarach, W. Sirichotedumrong, S. Nonesung, T. Jaknamon, P. Pengpun, P. Taveekitworachai, A. Na-Thalang, S. Sripaisarnmongkol, K. Jirayoot, and K. Tharnpipitchai, “Typhoon 2: A family of open text and multimodal thai large language models,” 2024.
- [12] D. Kim, S. Lee, Y. Kim, A. Rutherford, and C. Park, “Representing the under-represented: Cultural and core capability benchmarks for developing thai large language models,” in *Proceedings of the 31st International Conference on Computational Linguistics*, 2025, pp. 4114–4129. [Online]. Available: <https://aclanthology.org/2025.coling-main.278/>
- [13] L. P. Argyle, E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate, “Out of one, many: Using language models to simulate human samples,” *Political Analysis*, vol. 31, no. 3, pp. 337–351, 2023.
- [14] J. Mooney, J. Woldense, Z. R. Jia, S. A. Hayati, M. H. Nguyen, V. Raheja, and D. Kang, “Are LLM agents behaviorally coherent? latent profiles for social simulation,” 2025.
- [15] N. Schwager, S. Munker, A. Plum, and A. Rettinger, “Towards simulating social media users with LLMs: Evaluating the operational validity of conditioned comment prediction,” in *Proceedings of the 15th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 2026, pp. 208–221. [Online]. Available: <https://aclanthology.org/2026.wassa-1.16/>
- [16] L. Cross, N. Haber, and D. L. K. Yamins, “Validating generative agent-based models of social norm enforcement: From replication to novel predictions,” 2025.
- [17] D. Hazarika, S. Poria, S. Gorantla, E. Cambria, R. Zimmermann, and R. Mihalcea, “CASCADE: Contextual sarcasm detection in online discussion forums,” in *Proceedings of the 27th International Conference on Computational Linguistics*, 2018, pp. 1837–1848. [Online]. Available: <https://aclanthology.org/C18-1156/>
- [18] Apify, “Actors documentation: Runs and builds,” Online documentation, 2026, accessed 12 August 2026. [Online]. Available: <https://docs.apify.com/actors/running/runs-and-builds>
- [19] ISAL-NLP Laboratory, “MarketFish: A dual-llm agent-based social media simulation engine with algorithmic seeding,” 2026, unpublished architecture manuscript.
- [20] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT,” in *International Conference on Learning Representations*, 2020. [Online]. Available: <https://openreview.net/forum?id=SkeHuCVFDr>
- [21] OpenRouter, “Z.ai: GLM 5.3 Flash model card and api information,” Online, 2026, accessed 31 August 2026. [Online]. Available: <https://openrouter.ai/z-ai/glm-5.3-flash>
- [22] L. Shi, C. Ma, W. Liang, W. Ma, and S. Vosoughi, “Judging the judges: A systematic study of position bias in LLM-as-a-judge,” 2024.